ANSWER SHEET

EE3014/EE3042 Final Exam Duration: 75 minutes Date: 26 January, 2024

Student Name and Number

The notes and books will be kept closed. The calculator can be used. All necessary formulas are given at the end of the page.

Part A

1) (25p) A 100-MVA, 232/115-kV, $\Delta - Y$ three-phase power transformer has a per-unit resistance of 0.018 pu and a per-unit reactance of 0.07 pu. The excitation branch elements are $Rc = 98$pu and $X_M = 21$pu.

a) If this transformer supplies a load of 78 MVA at 0.88 PF lagging, draw the phasor diagram of one phase of the transformer.

b) What is the voltage regulation of the transformer bank under these conditions?

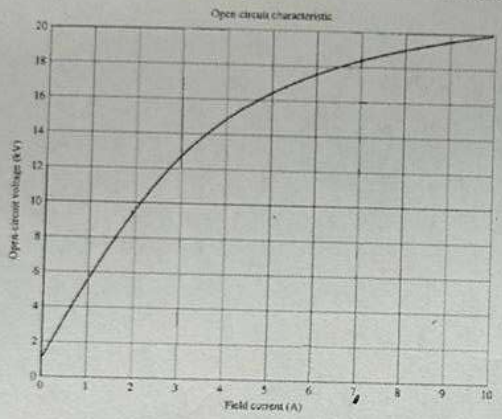
(30p) c) Sketch the equivalent circuit referred to the low-voltage side of one phase of this transformer.

2) A 13.8-kV, 50-MVA, 0.85-power-factor-lagging, 50-Hz, four-pole, and wye-connected synchronous generator has a synchronous reactance of 2.5 Ω and an armature resistance of 0.2 Ω. At 50 Hz, its friction and windage losses are 1.1 MW, and its core losses are 1.45 MW. The field circuit has a dc voltage of 120 V, and the maximum $I_F = 9$ A. The current of the field circuit is adjustable over the range from 0 to 9 A. The open circuit characteristic curve (OCC) for the generator is given as shown in the figure. Assume that the field current of the generator has been adjusted to value of 4.6 A.

a) What will the terminal voltage of this generator be if it is connected to a wye-connected load with an impedance of 23 $\angle 25°$?

b) Sketch the phasor diagram of this generator.

c) What is the efficiency of the generator at these conditions?



(25p)

3) A 480-V, 102-kW, 0.82-pf-leading, 50-Hz, four-pole, Y-connected synchronous motor has a synchronous reactance of 1.8 Ohms and a negligible armature resistance. The rotational losses are also to be ignored. This motor is to be operated over a continuous range of speeds from 300 to 1500 r/min, where the speed changes are to be accomplished by controlling the system frequency with a solid-state drive.

a) Over what range the input frequency must be varied to provide this speed control range?

b) How large is E_A at the motor's rated conditions?

c) What is the maximum power the motor can produce at rated speed with the E_A calculated in part (b)?

d) What is the largest value that E_A could be at 300 r/min?

4) CANCEL A three-phase Y-connected 220-V (line-to-line) 7.55-kW 50-Hz six-pole induction motor has the following parameter values in Ω/phase referred to the stator: For a slip of 1.12 %, compute

a) The speed,

b) The load torque and output power

c) The stator current,

d) The power factor,

e) The efficiency when the motor is operated at rated voltage and frequency.

Part B (20p)

1) **What will happen if the relative speed between the rotating flux of stator and the rotor of the induction motor is zero?**

a) The slip of the motor will be 5%

b) The rotor will not run.

c) The rotor will run at very high speed.

d) The torque produced will be very large.

e) The slip of the motor will not be constant.

2) **It is necessary to provide 1000 kW of 60-Hz power at a location in Europe. The only power sources available operate at 50 Hz. It is decided to generate the power by means of a motor-generator set consisting of a synchronous motor**

driving a synchronous generator. How many poles should each of the two machines have to convert 50-Hz power to 60-Hz power?

a) 12/10
b) 6/4
c) 8/6
d) 8/4
e) 10/8

3) **The starting torque of a three-phase induction motor can be increased by**

a) Increasing the slip
b) Increasing the current
c) Both a) and b)
d) None of them
e) Decreasing the slip and increasing the current

4) **A 480-V, 102-kW, 0.82-pf-leading, 50-Hz, four-pole, Y-connected synchronous motor has a synchronous reactance of 1.8 Ohms and a negligible armature resistance. The rotational losses are also to be ignored. This motor is to be operated over a continuous range of speeds from 300 to 1500 r/min, where the speed changes are to be accomplished by controlling the system frequency with a solid-state drive. Over what range the input frequency must be varied to provide this speed control range?**

a) [20-100] Hz
b) [50-60] Hz
c) [10-60] Hz
d) [20-60] Hz
e) [10-50] Hz

5) **What will happen if the field current of oncoming generator on a large power system increased?**

a) The terminal voltage increases.
b) The frequency of the system increases.
c) The armature voltage of the generator increases.
d) The oncoming generator will work as a motor.
e) Both b) and c)

6) **How can the reactive power sharing between two generators be controlled without affecting the system's terminal voltage?**

a) No way to control the reactive power sharing two generators without affecting the system's terminal voltage.
b) By simultaneously reducing the field current of the more reactive power generator and increasing the field current with the lower reactive power generator.
c) By simultaneously reducing the governor set point of the more reactive power generator and increasing the governor set point with the lower reactive power generator.
d) By simultaneously increasing the field current of the more reactive power generator and decreasing the field current with the lower reactive power generator.
e) By simultaneously increasing the governor set point of the more reactive power generator and decreasing the governor set point with the lower reactive power generator.

7) **Reducing the load power of lagging power factor synchronous motor**

a) Increases the power factor angle of synchronous motor.
b) Reduces the power factor angle of synchronous motor.
c) Increases the input real power.
d) It will not change the power factor angle of synchronous motor.
e) Both a) and c)

8) **During the open circuit test of a transformer**

a) The secondary is supplied rated KVA.
b) Primary is supplied with no-load current.
c) Primary is supplied current at high voltage.
d) Primary is supplied rated voltage.
e) Both b) and d)

$P_{in} = \sqrt{3} V_T I_L \cos\theta$	$Q = V_s I_s \sin\theta$ per phase	$P = 3 V_s I_s \cos\theta$
$n_{sync} = \frac{120f}{p}$	$s = \frac{n_{sync} - n_{rotor}}{n_{sync}}$	$\tau_{ind} = \frac{P_{AG}}{w_{sync}}$
$P_{out} = P_{conv} - P_{F\&W} - P_{misc}$	$P_{conv} = P_{AG} - P_{RCL}$	$P = \frac{3 V_\phi E_A \sin\delta}{X_S}$

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1) A 100-MVA, 232/115-kV, Δ-Y 3-Ø Power transformer (25)

$R_{eq} = 0{,}0018$ pu, $X_{eq} = 0{,}07$ pu

The excitation branch: $R_c = 98$ pu, $X_M = 21$ pu.

a) Load: 78 MVA at 0,88 pf.

Phasor diagram?

b) VR?

c) Equivalent circuit referred LVS of the transformer.

a) $I_{L,s} = \dfrac{S}{\sqrt{3}\, V_{L,s}} = \dfrac{78\,MVA}{\sqrt{3}(115\,kV)} = 391.6\,A$ (load side line current) 2/

$I_{L,s\,base} = \dfrac{S_{base}}{\sqrt{3}\, V_{L,base}} = \dfrac{100\,MVA}{\sqrt{3}(115\,kV)} = 502\,A$ 2/

$I_{L,s,pu} = \dfrac{I_{L,s}}{I_{L,base}} = \dfrac{391.6}{502} = 0{,}78$ 1/ , $pf = 0{,}88$ lagging $\Rightarrow \theta = 28.36°$ 1/

$\Rightarrow \vec{I}_{L,s,pu} = 0{,}78\angle -28{,}36°$ 1/

The transformer referred to the secondary side. Hence,

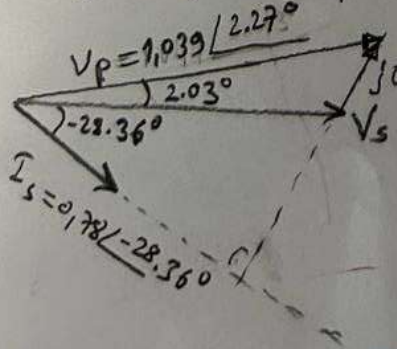
$V_{s,pu} = 1\angle 0° \Rightarrow V_{p,pu} = 1\angle 0° + 0{,}78\angle -28.36° (Z_{eq,pu})$

$Z_{eq,pu} = R_{eq} + jX_{eq} = 0{,}018 + j0{,}07 = 0{,}072\angle 75{,}58°$ pu 2/

$\Rightarrow V_{p,pu} = 1\angle 0° + 0{,}78\angle -28{,}36° (0{,}072\angle 75{,}58°) = 1{,}039\angle 2.27°$ pu. 3/

$jX_{eq,pu}\, I_{L,s,pu} = j(0{,}07)(0{,}78) = j0{,}055$ pu

Ignoring R_{eq} in the phasor diagram.



2/

1) continued

b) $$VR = \frac{V_{P,pu} - V_{S,pu}}{V_{S,pu}} \times 100 = \frac{1.039 - 1.0}{1.0} \times 100 = 3.9\ \%$$ 3/

c) per phase Low voltage side (LVS) equivalent circuit.

$$V_{\phi,S,base} = \frac{V_{L,S,base}}{\sqrt{3}} = \frac{115kV}{\sqrt{3}} = 66,36kV \ (Y)$$ 1/

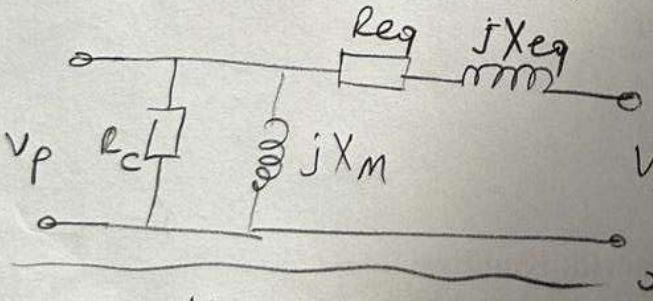
$$Z_{base} = \frac{V_{\phi,S,base}}{I_{L,S}} = \frac{66,36kV}{502A} = 132.2\ \Omega$$ 2/

$$R_{eq,S} = (R_{eq,pu})(Z_{base}) = (0,0018)(132,2) = 0,238\ \Omega$$ 1/

$$X_{eq,S} = (X_{eq,pu})(Z_{base}) = (0,07)(132,2) = 9,254\ \Omega$$ 1/

$$R_c = (R_{c,pu})(Z_{base}) = (98)(132,2) \cong 13k$$ 1/

$$X_M = (X_{M,pu})(Z_{base}) = (21)(132,2) \cong 2,8k$$ 1/



1/

2) A 13,8-kV, 50-MVA, pf = 0,85 Lagging, 50 Hz, 4-pole, Y-connected (30)

SG: $X_S = 2,5\ \Omega$, $R_A = 0,2\ \Omega$, At f = 50 Hz $P_{Few} = 1.1MW$, $P_{core} = 1.45MW$

$V_F = 120\ V\ DC$

$I_{F_{max}} = 9A$ If range: 0 – 9A

$I_{F_{adjusted}} = 4,6A$

a) $V_T = ?$ Y-connected load with $Z_L = 23\angle 25^\circ$

b) Sketch the phasor diagram of the SG.

c) $\eta_{SG} = ?$

3) 480V, 102kW, 0,82 pf-leading, 50Hz, 4-pole Y-connected SM (25)

$X_S = 1,7\,\Omega$, $R_A \approx 0$ $P_{rot} \approx 0$, range of speed: [300-1500] r/min

a) f_s → variation (range) ?

b) E_A = ? at rated conditions

c) $P_{o_{max}}$ = ? at rated speed with E_A (part b)

d) $E_{A_{max}}$ = ? at 300 r/min

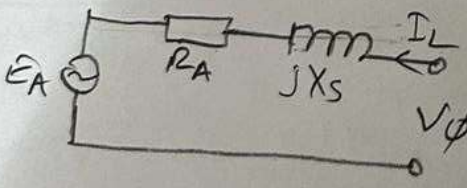
a) $f_{se,min} = \frac{(p)n_m}{120} = \frac{4(300\ r/min)}{120} = 10Hz$ 2/

$f_{se,max} = \frac{4(1500\ r/min)}{120} = 50Hz$ 2/

The freq. range: [10 – 50Hz] 1/

b) $V\phi = \frac{V_T}{\sqrt{3}} = \frac{480V}{\sqrt{3}} = 277.13V$ 1/

$I_A = I_L = \frac{P_{in}}{\sqrt{3}V_T \cos\theta} = \frac{102kW}{\sqrt{3}(480)(0,82)} = 149.62A$ 3/



$\theta = \cos^{-1}(0,82) \cong 35°$ 1/

$E_A = V\phi - (R_A + jX_S)I_A$

$E_A = 277.13\angle 0° - (1,7\angle 90°)(149.62\angle +35°)$

$E_A = 471.5\angle -26.2°$ 7/

c) $P_{o_{max}} = \frac{3V\phi E_A}{X_S} = \frac{(3)(277.13)(471.5)}{1,7} \cong 231kW$ 3/

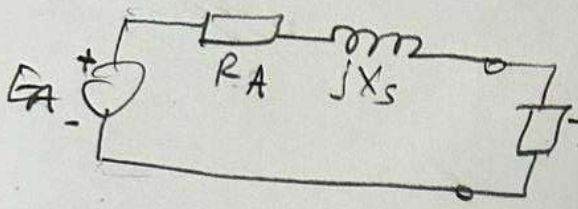
d) $E_{A,300\ r/min,\ max} = \frac{f_{se,300\ r/min}}{f_{se,1500\ r/min}} E_{A,rated} = \frac{300}{1500}(471.5) = 94.3V$

(max speed)

5/

6/6

2) continued From the OCC $V_T \cong 16\,kV$ for $I_F = 4.6\,A$ 2/



$\Rightarrow V\phi = E_A = \frac{V_T}{\sqrt{3}} = \frac{16\,kV}{\sqrt{3}}$ 2/ 1/

$V\phi = 9.24\,kV$ 1/

When the load is connected to the terminal of the SG.

$E_A = (R_A + jX_S + Z_L)\,I_A = (0,2 + j2.5 + 23\angle 25°)\,I_A$ 2/

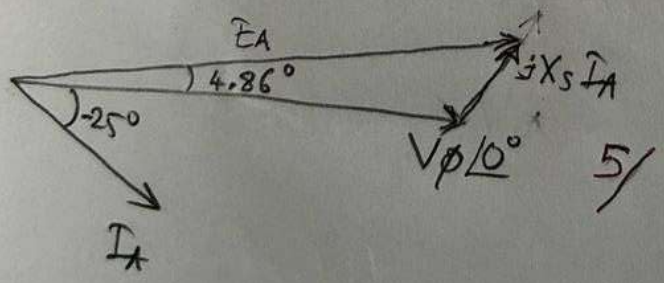
$$I_A = \frac{|E_A|}{|0,2 + j2.5 + \underbrace{20.85 + j9,72}_{Z_L}|} = \frac{9.24\,kV}{24.34\sqrt{2}} = 379.6\,A$$ 5/

$V\phi = Z_L I_A \Rightarrow V_T = \sqrt{3}\,V\phi = \sqrt{3}\,[(23)(379.6)] \cong 15\,kV$ 1/ 2/

b) $I_A = 379.6\angle -25°$, $V\phi = 9.24\angle 0°\,kV$ 2/ 1/

$E_A = V\phi + (R_A + jX_S)\,I_A = 9.24\angle 0°\,kV + \underbrace{(0,2 + j2.5)}_{2.51\angle 85.43°}(379.6\angle -25°)$ $\underbrace{\qquad\qquad}_{949\angle 60.43}$

$E_A = 9743.3\angle 4.86°$ 6/



5/

5/6

c) $P_{in} = P_{FLW} + P_{core} + P_{out} \Rightarrow P_{out} = \sqrt{3} V_T I_L \cos\theta = 3 V_\phi I_A \cos\theta$

$\Rightarrow P_{out} = (3)(9.240)(379.6)\cos(-25) = 9.54\,MW$ 1/

$P_{in} = 1.1\,MW + 1.45\,MW + 9.54\,MW = 12.1\,MW$ 1/

$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{9.54}{12.1} \times 100 = 78.8\%$ 1/

5/6